

Quality of Research Effort Rubric

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Grading System

This rubric uses a **specifications grading** system. Assignments are evaluated on a **Complete / Incomplete** basis against a clear set of requirements. To earn a specific final grade, a student must complete "bundles" of work that increase in complexity and rigor.

Grade Bundles

Grade	Requirements
A	Meets all requirements for a B , plus the Advanced Analysis specifications.
B	Meets all requirements for a C , plus the Data Integration & Visualization specifications.
C	Meets all Foundational Code & Documentation specifications.
F	Fails to meet the foundational specifications for a C .

Technical Specifications

1. Foundational Code & Documentation — *The C Level*

- **Functional Integrity:** The analysis pipeline executes without errors on the primary dataset. At minimum, this includes data loading, quality control, normalization, dimensionality reduction, and clustering.
- **Version Control:** The project includes a Git repository with at least **five meaningful commits** that document the development process (not just one large initial commit).
- **Documentation:** A progress or summary document is present and kept up to date. It must include: a plain-language description of the project goals, a summary of datasets used (e.g., accession numbers, cell counts, conditions), and a record of key findings at each stage.

- **Code Style:** Code follows standard conventions (e.g., PEP 8 for Python). Notebooks or scripts are organized in logical sections with clear headers. Variables and functions are named descriptively.
 - **Core Analysis:** Successfully performs the standard scRNA-seq preprocessing workflow, including:
 - Quality control filtering (e.g., mitochondrial content, gene count thresholds)
 - Normalization and dimensionality reduction (e.g., PCA, UMAP)
 - Clustering and basic cell population identification using known marker genes
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2. Data Integration & Visualization — *The B Level*

- **Error Handling:** The pipeline includes basic checks to handle common issues gracefully (e.g., missing files, mismatched gene names between datasets, or failed downloads) with informative messages rather than silent crashes.
 - **Multi-dataset Integration:** Successfully loads and integrates at least one additional dataset alongside the primary dataset (e.g., a published reference dataset used for cross-validation or label transfer). Batch correction is applied where appropriate (e.g., Harmony, scVI, or equivalent).
 - **Visualization:** Produces at least **three high-quality, labeled figures** exported as .png or .pdf, covering different aspects of the analysis (e.g., cluster overview, marker gene expression, condition comparison, or population-specific patterns).
 - **Modularity:** The code is organized into well-named notebooks or scripts, each with a single clear purpose (e.g., QC, clustering, integration, visualization). Non-obvious steps are explained with brief comments.
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3. Advanced Analysis — *The A Level*

- **Non-trivial Biological Analysis:** Implements at least **one** analysis that goes beyond standard preprocessing and clustering, such as:
 - Transcription factor network inference (e.g., pySCENIC or equivalent)
 - Cell fate trajectory analysis (e.g., CellRank, Monocle, or equivalent)
 - Cell-cell communication analysis (e.g., LIANA, CellChat, or equivalent)
 - A supervised classifier to distinguish biological states (e.g., LASSO, XGBoost, or equivalent), validated on a held-out dataset
- **Statistical Rigor:** All differential expression or enrichment results include appropriate multiple testing correction (e.g., FDR / Benjamini-Hochberg). Where applicable, model performance is reported with proper validation metrics (e.g., AUC, cross-validation).

- **Reproducibility:** The project includes an environment specification file (e.g., environment.yml for Conda, or requirements.txt) that allows another user to recreate the software environment and reproduce all results.

Professionalism & Engagement (Updated Grade Bundles)

Grade	Technical Requirements	Engagement & Lab Citizenship
A	Meets all A technical specs	Meets all B specs, plus Leadership/Initiative criteria
B	Meets all B technical specs	Meets all C specs, plus Active Participation criteria
C	Meets all C technical specs	Meets all Foundational Reliability criteria

Engagement Specifications

1. Foundational Reliability — *The C Level*

- **Core Hours:** Maintains a consistent presence during agreed-upon core hours (e.g., 10:00 AM – 3:00 PM) to ensure availability for collaboration and troubleshooting.
 - **Attendance:** Attends at least **90%** of scheduled lab meetings. Arrives on time and stays for the duration.
 - **Communication:** Responds to lab-related communications (e.g., Slack, email) within **one business day**.
 - **Documentation:** Updates the shared lab notebook or project board whenever a meaningful milestone is reached (e.g., after completing a new analysis step, resolving a major issue, or obtaining a new dataset).
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2. Active Participation — *The B Level*

- **Lab Meeting Contribution:** Provides regular status updates during stand-ups and engages constructively during other members' presentations (e.g., asking questions, offering suggestions).
- **Peer Feedback:** Provides thoughtful written feedback on at least **two** lab mates' documents, protocols, or analysis outputs during the project period.

- **Troubleshooting Independence:** Demonstrates a "Search, Then Ask" workflow — exhausts available documentation, forums, and resources before escalating to a senior lab member.
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3. Leadership & Initiative — *The A Level*

- **Onboarding / Resource Creation:** Creates at least one resource that will benefit future students or lab members — for example, a quick-start guide, a troubleshooting FAQ, or a well-documented template notebook for a tool used in the project.
 - **Adaptive Scheduling:** Proactively identifies and communicates upcoming bottlenecks (e.g., large data downloads, compute-heavy steps, or external dependencies) at least one week in advance to allow for planning.
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Grading Key

Bundle Completed	Final Grade
Bundle A	90+
Bundle B	80–90
Bundle C	70–80
Does not meet Bundle C	< 70